SAS CONTROL MODULE [TWO-STEP DEPLOYMENT CONTROL SYSTEM]

id0810b1500200

Purpose
- The SAS control module controls the air bag system.

Function
Air bag system initialization diagnostics function
- When the ignition is switched ON, the SAS control module turns on the air bag/seat belt pre-tensioner system warning light in the instrument cluster for approx. **6 s** to indicate that the air bag system is performing initialization diagnostics.
- If there is no malfunction with the air bag system, the air bag/seat belt pre-tensioner system warning light turns off.

Backup power supply function
- The condenser in the SAS control module discharges even if the power supply to the SAS control module has been cut off due to a collision, and the backup power supply function operates the air bag system for a specified period of time.

Passenger air bag deactivation (PAD) system initialization diagnostic function (With PAD system)
- When the ignition is switched ON, the SAS control module turns on the PAD indicator in the indicator unit for approx. **6 s** to indicate that the PAD system is performing initialization diagnostics.
- If there is no malfunction in the PAD system, the PAD indicator is turned off.

Vehicle travel conditions detection function
- The SAS control module detects the signals from the following integrated sensors.
- The SAS control module outputs the detection signals from the following sensors and outputs them to the DSC HU/CM via a CAN signal.

Sensor	Signal
Low-G sensor	• Vehicle longitudinal acceleration speed signal • Vehicle lateral acceleration speed signal
Yaw rate sensor	• Vehicle steering angle speed signal

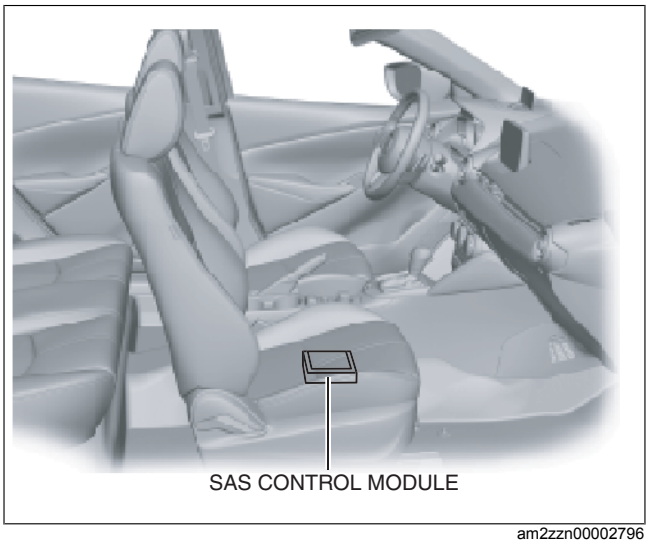
Collision detection signal output function
- The SAS control module sends the following signal to the following units when it determines that it is necessary for the air bag system to operate (deploy).
- When the signal is received, the each unit performs controls to mediate secondary injuries caused by the vehicle collision.

Related unit	Signal	Reference
• Body control module (BCM)	Secondary collision reduction signal	(See POWER DOOR LOCK SYSTEM.) (See EXTERIOR LIGHTING SYSTEM.)
• PCM • Instrument cluster • Center display	Fuel-cut request signal	(See FUEL INJECTION CONTROL SYSTEM [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See FUEL INJECTION AMOUNT CONTROL [SKYACTIV-D 1.5].) (See ENTERTAINMENT SYSTEM [WITH CENTER DISPLAY].)
• Electrical supply unit (ESU)	Collision detection unlock signal	(See POWER DOOR LOCK SYSTEM.)

Driver's seat position detection function
- The SAS control module detects the slide position (foremost position or position other than foremost position) of the driver's seat based on the signal from the seat position sensor. For details on the driver's seat position detection function, refer to the SEAT TRACK POSITION SENSOR. (See SEAT TRACK POSITION SENSOR [TWO-STEP DEPLOYMENT CONTROL SYSTEM].)

Construction
- The SAS control module is located underneath the rear console.
- The following sensors are integrated with the SAS control module.
  - — Determination and safing circuit
  - — Output control circuit
- The following sensors are integrated with SAS control module.
  - — Crash sensor

— Yaw rate sensor
— Low-G sensor



am2zzn00002796

Operation

- For the SAS control module operation, refer to the AIR BAG SYSTEM. (See AIR BAG SYSTEM [TWO-STEP DEPLOYMENT CONTROL SYSTEM].)

Fail-safe

- If a malfunction occurs in the air bag system and DTCs are detected, the SAS control module inhibits control of the air bag system so that they do not operate (deploy) accidentally, and it flashes the air bag/seat belt pre-tensioner system warning light.
- If a malfunction occurs in the seat track position sensor and DTCs are detected, the SAS control module controls the forward/back slide position of the driver's seat as the maximum forward position.
- If a malfunction occurs in the PAD system and DTCs are detected, the SAS control module performs control relevant to the PAD switch OFF status, and the PAD OFF indicator turns on.